

ARTHUR MLAMBO

Address: Rüdickenstraße 32, 13053, Berlin, DE
Phone: +49 17665964509
Email: arthurmlambo@gmail.com
LinkedIn: www.linkedin.com/in/art-mla | GitHub: github.com/art-mla



PROFILE

Data Science graduate (M.Sc.) and Mechatronics Engineer (B.Eng) with over five years of experience in industrial systems. Combines technical field expertise with advanced analytical skills in Python, SQL, and predictive analytics. Seeking a junior role to apply a hybrid engineering and data driven approach to solve real world challenges.

SKILLS

Languages & Frameworks: Python (PyTorch, TensorFlow/Keras, Scikit-Learn, Pandas, NumPy), SQL,

AI/ML: Supervised & unsupervised learning, time-series forecasting, anomaly detection, LLM concepts, agentic AI, feature engineering, model evaluation & backtesting

MLOps & Tools: FastAPI, Docker, GitHub Actions (CI/CD), MLflow, Apache Spark, dbt, AWS (S3/EC2), Git

Data & Visualisation: Power BI, Tableau, SQL pipelines, data exploration, stakeholder reporting

PROJECTS

Predictive Maintenance MLOps Pipeline

2025 – Present

Python | PyTorch | FastAPI | Docker | GitHub Actions | MLflow | SQL

- Building an end-to-end ML pipeline for fault prediction on time-series sensor data. Covers ingestion, feature engineering, training, and a FastAPI endpoint for model serving, containerised with Docker and deployed through a GitHub Actions CI/CD workflow.
- Added automated drift detection and retraining triggers so the model stays accurate without manual intervention.

M.Sc. Thesis: Hybrid ML for Predictive Maintenance

2025 – 2026

PyTorch | TensorFlow | Scikit-Learn | Pandas | AWS

- Compared classical time-series models against deep learning architectures for industrial fault prediction. Ran backtests across multiple sensor datasets to measure whether model complexity translated into real accuracy gains.
- Developed a hybrid 1D-CNN + Random Forest predictive maintenance framework for industrial asset failure detection using the AI4I 2020 dataset, achieving 97.6% ROC-AUC for binary classification and 0.861 Macro F1 for multiclass fault diagnosis.

WORK EXPERIENCE

Go Solar (Pvt) Ltd., Cape Town

Jan 2023 – Jun 2024

Solar Systems Engineer

- Designed and supervised the installation of more than 100+ solar installations.
- Coordinated delivery across subcontractors and clients, which meant translating technical findings into language that non-engineers could act on.
- System Integration: Delivered 100+ industrial power systems, managing the full lifecycle from equipment delivery to grid integration.

- **Data Monitoring:** Implemented performance monitoring systems to reduce downtime and optimize system faults.
- **Stakeholder Management:** Coordinated cross-functional teams and subcontractors for global standard compliance.

Frogfoot Networks, Cape Town

Oct 2018 – Dec 2020

Field Systems & Network Technician

- **Network Infrastructure:** Executed field splicing and network fault localization in high-demand industrial environments.
- **Interface Management:** Resolved connectivity issues between hardware endpoints and central network switching.
- **Process Documentation:** Maintained technical logs and resolution records across a portfolio of clients sites.

Steel Makers (Pvt) Ltd.

Jan 2014 – Dec 2014

Electrical/Mechanical Engineering Intern

- **Industrial Control Systems & Data:** Worked with Siemens S7 PLC equipment in a steel manufacturing environment; gained early exposure to time-series control data, sensor fault patterns, and automated system behavior.
- **Fault Diagnosis & Corrective Action:** Diagnosed electrical and mechanical faults at the hardware/control software interface; performed structured root cause analysis and implemented corrective actions, developing the systematic problem.

EDUCATION

M.Sc. Data Science (Distinction)

Feb 2026

Arden University, Berlin

Modules: Machine Learning, Deep Learning, NLP, Big Data & Spark, Cloud Computing (AWS), AI Ethics, Statistical Modelling

B.Eng. (Hons) Mechatronics

Dec 2015

Chinhoyi University of Technology, Zimbabwe

Focus: Control systems, signal processing, data acquisition, embedded systems

LANGUAGES & CERTIFICATIONS

Languages: English — Native/Fluent (C2) | German — Basic (A2)

Memberships: Verein Deutscher Ingenieure (VDI), Berlin Chapter | PrTech — South African Institute of Engineers.